

* Design a combination circuit which converts 3 bits binary number into 2's complement using NAND gates.

Truth Table

(Inputs)			(Outputs)		
a	b	c	x	y	z
0	0	0	0	0	0
0	0	1	1	1	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	0
1	0	1	0	1	1
1	1	0	0	1	0
1	1	1	0	0	1

K-map and Boolean Equation

x

a \ bc	00	01	11	10
0		1	1	1
1	1			

y

a \ bc	00	01	11	10
0		1		1
1		1		1

$$\begin{aligned}
 x &= \bar{a}c + \bar{a}b + a\bar{b}\bar{c} \\
 &= \bar{a}c + \bar{a}b + a\bar{b}\bar{c} \\
 &= \bar{a}c \cdot \bar{a}b \cdot a\bar{b}\bar{c}
 \end{aligned}$$

$$\begin{aligned}
 y &= \bar{b}c + b\bar{c} \\
 &= \bar{b}c + b\bar{c} \\
 &= \bar{b}c \cdot b\bar{c}
 \end{aligned}$$

z

a \ bc	00	01	11	10
0		1	1	
1		1	1	

$$z = c$$

Logic Diagram

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